Swin-AQI: Depth-Aware Vision Transformer with Physics-Informed Learning for Particulate Matter Estimation from Ground Imagery

Van Truong Thinh Nguyen [1,2], Thanh-Hai Tran [1,2], and Xuan-Bach Le [1,2]⋆

[1] Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology (HCMUT), Ho Chi Minh City, Vietnam
[2] Vietnam National University Ho Chi Minh City, Ho Chi Minh City, Vietnam
{thinh.nguyen0334,tthai.sdh241,lexuanbach}@hcmut.edu.vn

Abstract. Street-level imagery is an appealing, low-cost complement to air-quality stations, but a camera is only useful when its pixels respond to the pollutant of interest. For particulate matter they do: aerosols scatter light along the line of sight, so distant buildings, sky boundaries, and horizon regions lose contrast as aerosol loading rises. *Swin-AQI* is a depth-aware Swin Transformer for monocular $PM_{2.5}$ and PM_{10} estimation from street-level urban images: a MiDaS-derived depth-coherence bias is applied to post-backbone attention, learnable Fourier features encode the capture hour, and Beer–Lambert and Koschmieder terms keep predictions consistent with visibility physics. Supervision is provided by SILAM atmospheric fields. On a dataset of 1,977 daylight images from Ho Chi Minh City, Swin-AQI reaches R^2=0.80 for $PM_{2.5}$ and R^2=0.76 for PM_{10} on a random held-out test split – a within-distribution result, since only 35 distinct label combinations span the whole dataset (Section 4.2) – outperforming every CNN and transformer baseline under a standard single-head MSE objective on the same images and labels. Ablations show hour-of-day and uncertainty weighting contribute the most, largely for the same reason, while depth-aware attention and the physics terms add smaller but separable gains. The scope is also deliberately constrained: Swin-AQI is a single-city, moderate-pollution estimator whose gaseous outputs stay contextual.

Keywords: Particulate matter estimation · Depth-aware vision transformer · Physics-informed learning · Street-level imagery

1 Introduction

Urban particulate exposure varies over short distances and poses documented health risks [12,17,27]. Reference stations are sparse, and low-cost sensors require calibration and drift management [16,21]. Street-level imagery offers a complementary signal because particulate matter changes image formation: aerosols

⋆ Corresponding author

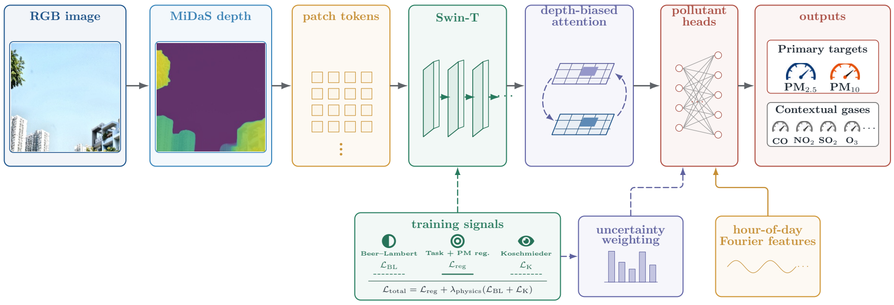

Fig. 1: Swin-AQI framework: a depth-coherence attention bias (B_{depth}) is applied to post-Swin features and fused with the encoded capture time; the heads are trained with uncertainty-weighted regression ($\mathcal{L}_{\mathrm{reg}}$, combining smooth-L1 tasks with PM-ordering) and weak physics losses ($\mathcal{L}_{\mathrm{BL}}$, $\mathcal{L}_{\mathrm{K}}$).

scatter light along the line of sight, reducing far-field contrast, obscuring distant structure, and weakening sky boundaries [1,11]. A visual estimator should therefore be depth-aware, grouping tokens at similar optical depth to pool haze evidence coherently. *Swin-AQI* is proposed to meet this need, previewed below.

Existing image-based methods get part of the way there. Hand-crafted visibility features exploit contrast, edge strength, and sky appearance, but remain sensitive to illumination and scene layout [13]. Convolutional regressors improve robustness, yet typically learn a generic image-to-pollution mapping without explicitly privileging path-length-dependent haze evidence [28,29].

Figure 1 presents *Swin-AQI*, a depth-aware Swin Transformer for estimating $PM_{2.5}$ and PM_{10} from ground imagery that addresses the modeling gaps identified above. The model preserves the hierarchical shifted-window design of Swin [14] but appends a depth-coherence attention module over post-backbone features, encoding path-length-dependent haze structure. It further incorporates hour-of-day Fourier features and weak Beer–Lambert/Koschmieder consistency terms, and retains the gaseous outputs only as auxiliary contextual tasks rather than as primary RGB-sensing claims.

This study makes four main contributions. First, a Ho Chi Minh City street-image dataset of 1,977 street-level daylight images paired with hourly SILAM fields is assembled. Second, a post-backbone depth-coherence attention bias is introduced to group tokens at similar optical depths. Third, a scoped training objective is formulated with hour-of-day encoding, uncertainty-weighted multi-task regression, and weak visibility-consistency losses. Fourth, Swin-AQI is compared against modern CNN and transformer baselines trained under a standard single-head, unweighted-MSE objective on the same images and labels, and its individual components are separately isolated via ablation.

The remainder of the paper is organized as follows. Section 2 details the Swin-AQI architecture, its depth-aware attention bias, and the time-encoding and physics-informed training objective. Section 3 describes the dataset, supervision, and baseline protocol, and Section 4 reports the comparison against CNN and transformer baselines together with ablations. Section 5 discusses diagnostics, scope, and limitations, Section 6 situates the proposed approach within prior work, Section 7 concludes, and Section 8 outlines future work.

2 Methodology

Figure 1 walks through Swin-AQI from input to prediction: the solid path is inference, while the dashed paths are the weak training signals that nudge particulate predictions toward physics without ever replacing the SILAM supervision. Section 2 of the supplementary material gives the same computation as pseudocode. Particulate matter ($PM_{2.5}$ and PM_{10}) is treated as the primary output, since distance-dependent haze gives them a physically grounded image cue. The four gases CO, NO_2, SO_2, and O_3 serve as auxiliary context and are interpreted solely in that role.

2.1 Depth-Aware Visual Encoding

Within each shifted window, standard Swin attention computes

$$\mathrm{Attn}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}} + B_{\mathrm{pos}}\right)V, \tag{1}$$

where d_k is the key dimension (kept distinct from the Beer–Lambert path length d_{km} introduced below). Standard attention treats near and far tokens alike once they share a window. However, because haze is depth-dependent, tokens at similar optical depths share more correlated particulate information than those at different depths. A relative depth map $D \in \mathbb{R}^{H\times W}$ is estimated with a frozen DPT-Large depth estimator [19] (the MiDaS v3.0 model [20], referred to as MiDaS below). Depth is average-pooled to the post-Swin token grid (N_{tok}=49 at 224×224) and min–max rescaled to $[0,1]$ per sample, and for every token pair $(i,\,j)$, a pairwise depth-similarity bias is defined:

$$\hat{D}_i = \frac{D_i - \min_k D_k}{\max_k D_k - \min_k D_k}, \qquad B_{\mathrm{depth}}(i,j) = \frac{0.1}{\tau}\,\exp\left(-\frac{|\hat{D}_i - \hat{D}_j|}{\tau}\right), \tag{2}$$

where $\hat{D}_i \in [0,1]$ is the per-sample min–max rescaled depth of token i and τ is a single learnable scalar temperature (initialized to 0.1) that controls the sharpness of the depth-similarity decay; this initialization sets the bias prefactor to unity. Then, a depth-aware attention module is applied to the post-Swin features $\mathbf{F} \in \mathbb{R}^{B\times N_{\mathrm{tok}}\times C}$. The attention update over the post-backbone features becomes

$$\mathrm{Attn}_{\mathrm{depth}}(\mathbf{F}) = \mathrm{softmax}\left(\frac{QK^{\top}}{\sqrt{d_h}} + B_{\mathrm{depth}}\right)V, \qquad [Q,\,K,\,V] = \mathbf{F}\,W_{QKV}, \tag{3}$$

where $B_{\text{depth}} \in \mathbb{R}^{N_{\text{tok}} \times N_{\text{tok}}}$ is broadcast identically across all attention heads and d_h is the per-head dimension. The bias does not override the learned image interactions, but encodes a depth-coherence prior. Token pairs at similar estimated depths receive a stronger additive attention bonus, while pairs separated by a large depth difference are downweighted, with the sharpness of that transition governed by τ.

2.2 Time Encoding and Prediction Heads

Urban particulate levels typically follow diurnal traffic, activity, and boundary-layer cycles, so hour-of-day supplies a complementary temporal prior. Because such a prior can also act as a shortcut when the visual signal is weak, capture time is treated as an auxiliary input rather than a substitute for image evidence. The capture hour $t \in \{0, \dots, 23\}$ is encoded with learnable Fourier features:

$$\theta_i(t) = 2\pi \left(f_i |s_i| \right) \frac{t}{24}, \qquad \phi(t) = \bigoplus_{i=1}^{N_{\text{freq}}} \left[\sin \theta_i(t), \cos \theta_i(t) \right]. \tag{4}$$

Here $f_i \in \{1, 2, 3, 4, 6, 8\}$ are fixed base frequencies, s_i are learned per-frequency scale parameters (initialized to 1), and N_{freq}=6. The encoding follows transformer positional encodings [26] and Fourier feature mappings [24]. The feature vector $\phi(t)$ is concatenated with the pooled depth-aware visual representation and passed to pollutant-specific MLP heads. The six outputs are trained with smooth L1 losses and homoscedastic task uncertainties σ_k^2 [10].

2.3 Training Objective

The training objective combines supervised regression ($\mathcal{L}_{\text{reg}}$), particulate ordering, and weak physics consistency:

$$\mathcal{L}_{\text{PM}} = \frac{1}{B} \sum_{i=1}^{B} \max\big(0,\ \hat{c}^{(i)}_{\text{PM}_{2.5}} - \hat{c}^{(i)}_{\text{PM}_{10}}\big), \tag{5}$$

$$\mathcal{L}_{\text{reg}} = \sum_{k=1}^{6} \left(\frac{1}{2\sigma_k^2} \mathcal{L}_k + \log(1 + \sigma_k) \right) + \lambda_{\text{PM}} \mathcal{L}_{\text{PM}}, \quad \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{reg}} + \lambda_{\text{physics}} \mathcal{L}_{\text{physics}}, \tag{6}$$

where $\mathcal{L}_k$ is the smooth L1 loss [5], and $\mathcal{L}_{\text{reg}}$ combines uncertainty-weighted tasks with PM-ordering. The PM term penalizes predictions violating $\text{PM}_{2.5} \leq \text{PM}_{10}$; λ_{PM}=10.0 is a fixed scalar penalty (activating rarely, as the dataset mostly satisfies the constraint) and λ_{physics}=1.49 is Optuna-tuned. The $\log(1 + \sigma_k)$ regularizer is a stabilized variant of Kendall et al. [10] to prevent collapse.

The physics term regularizes predictions with image proxies rather than calibrated radiometric supervision: $\mathcal{L}_{\text{physics}} = \mathcal{L}_{\text{BL}} + \mathcal{L}_{\text{K}}$. A spectral extinction coefficient $\beta = \alpha_{\text{fine}} \hat{c}_{\text{PM}_{2.5}} + \alpha_{\text{coarse}} m_{\text{coarse}} + \beta_{\text{Rayleigh}}$ is computed from the particulate predictions, with coarse mass $m_{\text{coarse}} = \max(\hat{c}_{\text{PM}_{10}} - \hat{c}_{\text{PM}_{2.5}}, 0)$, following the particulate structure of the modified IMPROVE equation [6]; $\alpha_{\text{fine}} = 3.0$, $\alpha_{\text{coarse}} = 0.6$, and $\beta_{\text{Rayleigh}} = 10.0$ are fixed constants, not learned, and β stays in Mm^{-1} throughout. Predicted transmittance is $\hat{T} = \exp(-\beta \times 10^{-3} \times d_{\text{km}})$, using a scene-relative path length

$$d_{\text{km}} = \left\langle 1 - \frac{\text{disp}}{\max(\text{disp})} \right\rangle_{\text{spatial}} \times d_{\text{assumed}}, \tag{7}$$

where disp is the MiDaS inverse-depth (disparity) map, $\langle \cdot \rangle_{\text{spatial}}$ denotes the spatial mean, and $d_{\text{assumed}} = 1\,\text{km}$ is a fixed scene-scale hyperparameter. On the image side, a spatial mean brightness value $\tilde{V}$ (uncalibrated in absolute scale, sitting in $[0, 0.004]$ due to double normalization in the codebase; supplementary Section 5) is matched to $\hat{T}$ and, separately, to a Koschmieder visibility estimate $\hat{V} = \text{clip}(K/\beta \times 10^3/100,\, 0,\, 1)$ with $K = 3.0$ a fixed constant (representing a 5% contrast threshold):

$$\mathcal{L}_{\text{BL}} = \left\| \hat{T} - \tilde{V} \right\|_1, \qquad \mathcal{L}_{\text{K}} = \left\| \hat{V} - \tilde{V} \right\|_1. \tag{8}$$

Together, these terms keep the particulate predictions compatible with the observed contrast decay, while the SILAM labels remain the supervision target; Section 1 of the supplementary material expands the derivation.

3 Experiments

This section evaluates Swin-AQI on street-level urban imagery with SILAM labels. Backbone comparisons and ablations are then reported. The dataset and model-related scripts are publicly available.[3,4]

3.1 Dataset and Labels

The dataset contains 1,977 street-level daylight images from Ho Chi Minh City, Vietnam, collected during a two-week period in April 2026 within a 5 km-radius area. Each image has resolution 4272×2848 and is paired with capture time and SILAM atmospheric-composition fields [22]. SILAM provides hourly estimates of $PM_{2.5}$, PM_{10}, CO, NO_2, SO_2, and O_3, drawn here from the global operational product (v6.1) natively gridded at 20 km. Each image is labeled by nearest-neighbor lookup against the hourly field in space and time. Because the study area is small relative to grid spacing, images in the same cell and hour receive identical labels (quantified in the supplement). The 1,977 images are randomly split 70/15/15 into train/val/test sets (seed 42), so the test split draws from the same label pool as training.

The collection mainly covers a moderate particulate regime. Figure 2 shows that most observations fall between 15–55 $\mu g/m^3$ for $PM_{2.5}$ and 40–80 $\mu g/m^3$ for PM_{10}, with smaller clusters at higher concentrations. The dataset therefore tests common daylight urban conditions in the study area, not severe haze episodes or heavily polluted regions.

[3] Dataset: https://kaggle.com/datasets/thnhnguynvntrng/swin-aqi-dataset

[4] Code: https://github.com/ZleeMb0334/acivs2026_swin-aqi

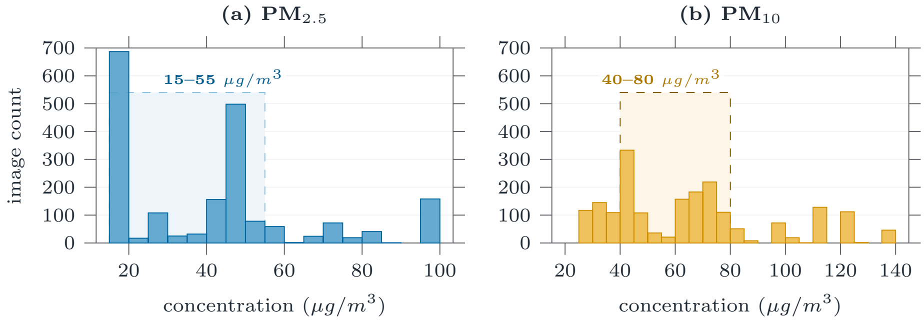

Fig. 2: Histograms of particulate matter observations.

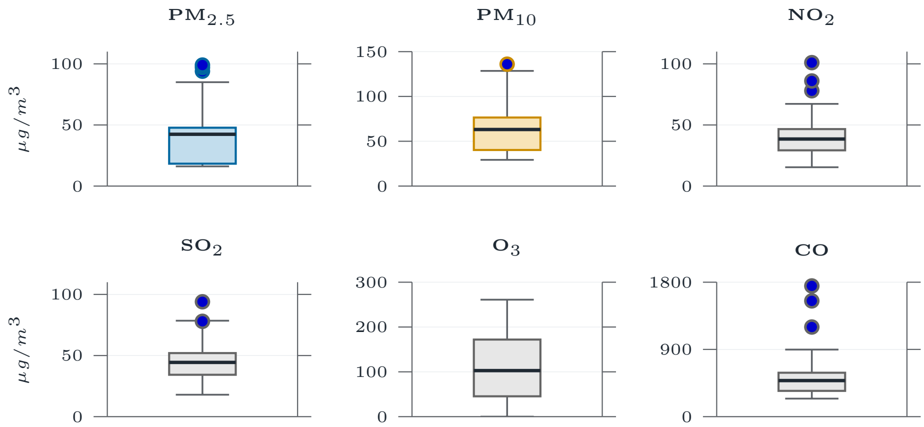

Fig. 3: Box plots of all six pollutants (concentrations in $\mu g/m^3$). Colored panels are the optical targets ($PM_{2.5}$, PM_{10}); gray panels are contextual gases.

Figure 3 places the particulate targets beside the four auxiliary gases. The wider CO scale and the quieter gas panels illustrate why the paper reports operational claims for $PM_{2.5}$ and PM_{10}, while treating gaseous outputs as contextual multi-task signals rather than stand-alone sensing results.

Images are resized to 224×224 for all models (including EfficientNet-B4, adapted from 380×380) and enhanced with CLAHE [30] (clipLimit=2.0, grid 8×8). Further augmentation is avoided, since aggressive transforms distort the haze and visibility cues the model relies on.

3.2 Baselines and Training

Performance is compared against ResNet-50 [8], EfficientNet-B4 [23], ConvNeXt-Tiny [15], ViT-Small [3], DeiT-Small [25], and standard Swin-T [14]. All models, including Swin-AQI, use ImageNet pretraining, AdamW, learning rate 10^{-4} with reduction on plateau, batch size 32 and early stopping.

Input fairness. All models receive the same resized image and the same six-pollutant supervision targets. The baselines do *not* receive the capture-hour feature, the uncertainty-weighted multi-task head, or the uncertainty-weighted loss: they use a plain linear regression head trained with unweighted MSE over the raw six-pollutant vector. This was chosen as a representative "standard practice" baseline configuration rather than a component-matched one.
Baseline scope. The baselines are modern general-purpose backbones rather than reimplementations of earlier image-to-AQI systems [13,28,29], which target different datasets and labels.

4 Results

This section addresses a focused evaluation question: once every model receives the same image, do the depth-aware and physics-informed components provide measurable performance gains? The results are presented in three tables: the first measures end-to-end accuracy against baselines, the second reports computational cost, and the third provides a component-wise ablation of Swin-AQI. Throughout, mean absolute error (MAE) and R^2 are reported.

4.1 Baseline comparisons

Table 1 compares Swin-AQI against baselines trained with a standard regression head and MSE loss, without the capture-hour feature or uncertainty weighting. Swin-AQI achieves superior performance (R^2=0.80 for $PM_{2.5}$, 0.76 for PM_{10}). The contrast with standard Swin-T is informative: both share a backbone, but the baseline lacks the capture-hour feature, uncertainty weighting, depth-aware attention, and physics loss.

Table 2 compares computational cost (measured on an RTX 2050 GPU and Core i5-13420H CPU). Swin-AQI has 375.7M parameters (31.6M trainable) and incurs 4.5 GFLOPs excluding the frozen MiDaS estimator. This causes higher latency and memory footprint (GPU latency roughly 7×, memory over 10× baselines), dominated by MiDaS requiring a full forward pass at inference.

4.2 Ablation

In Table 3, individual components are removed sequentially: depth-aware attention, time embedding, uncertainty weighting, and physics loss. To ensure a consistent comparison, every variant, including the full reference model, is evaluated under a fixed 50-epoch budget with identical hyperparameters. This budget accounts for why the "None (Full Swin-AQI)" row sits slightly below the fully trained model in Table 1 (0.79 vs. 0.80 for $PM_{2.5}$), making relative drops the primary evaluation metric.

The most significant decrease occurs when removing the time embedding: R^2 falls by 0.21 for $PM_{2.5}$ and 0.23 for PM_{10}. Uncertainty weighting matters almost as much, especially for $PM_{2.5}$. Depth-aware attention adds a further 0.10/0.10,

Table 1: Performance comparison under the same split and image input (baseline configuration described in the text; auxiliary gases are macro-averaged). Higher R^2 and lower MAE are preferred; best value per column in **bold**. All reported metrics are from single training runs.

	$\mathbf{PM_{2.5}}$		$\mathbf{PM_{10}}$		**Aux gases**
Model	$R^2\uparrow$	**MAE**$\downarrow$	$R^2\uparrow$	**MAE**$\downarrow$	$R^2\uparrow$
ConvNeXt-Tiny	0.27	16.15	0.32	18.57	0.18
EfficientNet-B4	0.38	13.99	0.40	16.40	0.44
ResNet-50	0.56	12.35	0.54	14.31	0.49
ViT-Small	0.22	16.78	0.25	20.22	0.11
Swin-T	0.24	16.30	0.27	18.86	0.14
DeiT-Small	0.31	14.12	0.28	17.80	0.29
Swin-AQI	**0.80**	**4.95**	**0.76**	**7.11**	**0.71**

Table 2: Computational cost. Latency on a single 224×224 image.

Model	Params	Trainable	GFLOPs	CPU (ms)	GPU (ms)	VRAM (MB)
ConvNeXt-Tiny	28.0	28.0	4.5	53.9 ± 4.7	9.6 ± 0.5	131
EfficientNet-B4	18.0	18.0	1.6	72.3 ± 2.0	12.0 ± 2.6	94
ResNet-50	24.0	24.0	4.1	61.9 ± 1.6	7.0 ± 0.6	112
ViT-Small	21.8	21.8	4.2	46.7 ± 3.3	8.2 ± 0.5	96
Swin-T	27.7	27.7	4.5	83.0 ± 4.3	10.0 ± 1.0	134
DeiT-Small	21.8	21.8	4.2	47.9 ± 1.6	8.1 ± 0.5	96
Swin-AQI	375.7	31.6	4.5*	553.0 ± 7.3	82.7 ± 0.8	1507

*Excludes frozen MiDaS backbone (∼61 GFLOPs).

and the physics loss buys a smaller particulate gain while accounting for a drop on the contextual gases (an unexplained multi-task interaction).

Swin-AQI operates beyond visual-only estimation: hour-of-day carries substantial predictive signal through diurnal traffic and meteorological cycles. Nevertheless, the "Time embedding" ablation row retains the depth-aware and physics mechanisms, and its R^2 drop of 0.21 exceeds the individual depth (0.10) and physics (0.07) drops. This drop must be interpreted with the label degeneracy analysis before it can be read as evidence of a genuine, complementary contribution.

This result should be read alongside the label degeneracy analysis in the supplementary material. Because the study area covers only 2 SILAM grid cells and 35 distinct (cell, hour) label combinations across 1,977 images, the label is essentially a function of the capture datetime-hour alone. The time-embedding ablation therefore dominates mechanically rather than demonstrating a learned visual cue.

Table 3: Ablation comparison under a shared 50-epoch budget (metric conventions as in Table 1; auxiliary gases macro-averaged). Drops from the full-reference row show each component's contribution. All reported metrics are from single training runs.

	$PM_{2.5}$		**PM_{10}**		**Aux gases**
Removed component	$R^2\uparrow$	**MAE**$\downarrow$	$R^2\uparrow$	**MAE**$\downarrow$	$R^2\uparrow$
None (Full Swin-AQI)	**0.79**	**6.07**	**0.74**	**8.53**	**0.70**
Depth-aware attention	0.69	7.36	0.64	9.94	0.49
Time embedding	0.58	8.73	0.51	11.74	0.44
Uncertainty weighting	0.60	8.81	0.57	11.33	0.59
Physics loss	0.72	6.90	0.69	9.27	0.36

5 Discussion

Swin-AQI performs best on daylight scenes where the particulate cue is visible: clear to moderately hazy weather, some distant structure or a sky boundary, and concentrations below the level where contrast loss saturates. In that regime, depth-aware attention matches image formation, since far-field contrast decay is a meaningful proxy for aerosol loading.

MiDaS gives relative, not metric, depth, so reflective glass, blank sky, dense vegetation, an unusual camera pitch, or rain on the lens can all distort the depth prior. The target labels come from the nearest SILAM grid cell and hour rather than per-image measurements, so a sharp local plume can be missed entirely if it falls outside the queried cell. And many everyday conditions, from fog and mist to backlighting, hard shadows, and exposure shifts, lower visibility for reasons that have nothing to do with particulates. Finally, $\tilde{V}$ operates as an uncalibrated architectural regularizer rather than a strict photometric constraint.

For these reasons, the operational scope remains focused: $PM_{2.5}$ and PM_{10} serve as a complementary sensing layer. The gaseous heads remain contextual, the dataset covers a single city at mostly moderate particulate levels, and the model relies in part on the capture hour. Immediate directions for future work include multi-city and seasonal evaluations, time-shuffled and held-out-hour controls, humidity-aware fog rejection, and selective fusion with calibrated sensors.

6 Related Work

Image-based air-quality estimation has used either hand-crafted visibility cues or learned image regression. Feature-based systems rely on contrast, edge strength, sky appearance, and related haze indicators [13], but these cues are sensitive to illumination, exposure, weather, and scene layout. Deep regressors reduce manual feature design [2,28,29], yet often treat the whole image as generic appearance evidence rather than modeling where particulate haze should be informative.

Atmospheric optics provide that missing structure. Beer–Lambert attenuation links extinction to path length, and Koschmieder visibility relates extinction to far-field contrast [1,11]. Dehazing methods exploit related assumptions to estimate transmission or recover a clearer image [4,7]. Swin-AQI repurposes this physical intuition for supervised particulate regression rather than image restoration.

Modern CNN and transformer backbones improve representation capacity, but backbone strength alone does not encode the path-length cue. CNNs learn hierarchical appearance patterns, and transformers learn token interactions [3,15]. Swin contributes efficient shifted-window attention [14], but standard attention has no built-in preference for path-length-correlated tokens. Swin-AQI instills a depth-coherence prior through a MiDaS relative-depth attention bias applied to post-backbone features.

The remaining components follow established learning principles. Sinusoidal positional encodings and Fourier feature mappings help neural networks represent periodic inputs [24,26], which motivates hour-of-day encoding. Physics-informed learning adds process-level constraints to supervised objectives [9,18], and uncertainty-weighted multi-task learning balances tasks with different noise levels [10]. Swin-AQI combines these ideas while keeping its operational claim limited to particulate matter.

7 Conclusion

Swin-AQI estimates particulate matter from ordinary street images by enabling a Swin-T backbone to incorporate distance-dependent scattering cues through relative-depth attention, hour-of-day Fourier features, multitask regression with uncertainty weighting, and weak Beer–Lambert/Koschmieder consistency terms in a unified architecture. On a single-city Ho Chi Minh City study, the model achieves R^2=0.80 for $PM_{2.5}$ and R^2=0.76 for PM_{10} on the test split (reflecting within-distribution fit under label degeneracy, as discussed in Section 4). Ablation results indicate that the temporal prior and uncertainty weighting contribute substantially to overall accuracy, while depth-aware attention and the physics loss provide additional separable gains. Camera imagery is thus presented as a dense complement to reference instruments under moderate daylight conditions—particularly valuable where monitoring stations are sparse—rather than a direct replacement.

8 Future Work

Future work will extend to multiple cities and seasons, validate SILAM labels with co-located sensors, and explore lightweight depth backbones to reduce the overhead in Table 2.

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Disclosure of Interests The authors have no competing interests to declare.

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